Modeling the spread of vaccination behavior and infection on survey-derived networks

Sophie Maretz, Juliana Taube, Shweta Bansal

Department of Biology

MOTIVATION

- Dual Contagion: Pathogens spread through physical proximity, while behaviors (like vaccination) may spread through social influence via technology and spatial proximity.
- Network Influence: Spatial and social interactions can be summarized using a **network**. Network structure dictates the speed and magnitude of both disease and social contagion.
- The U.S. Gap: here is limited data on higher-order network metrics (beyond simple degree) for the large, heterogeneous U.S. population.
- Epidemiological Risk: When groups of similar individuals are highly connected, infections can circulate within them, and protection (like vaccination) may not reach them as easily. Understanding these patterns helps us better assess risk and design interventions.

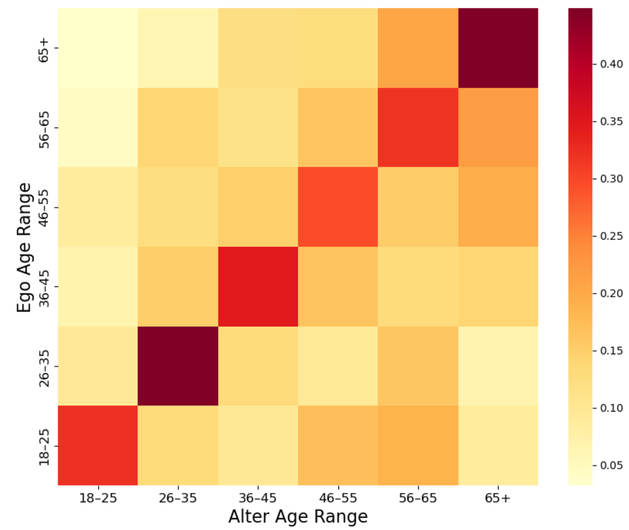
DATA

- RAND American Life Panel (February 2022) survey of 2,145 U.S. participants.
- Ego-Network Construction:
- Nodes: 2,145 egos (participants) and up to 15 alters (contacts) per person.
- Edges: Differentiated by contact type (Social vs. Spatial) and frequency.
- Demographics: Includes age, gender, race, urbanicity, and vaccination habits.

APPROACH

- Higher-Order Metrics: Calculate clustering and assortativity by age, gender, and vaccination status.
- Network Modeling: Use Exponential Random Graph Models (ERGMs) to make more complete social and spatial networks based on survey-derived patterns.
- Social Contagion Model: Test three behavioral mechanisms for vaccination uptake:
  - Simple Contagion: Requires only one exposure to spread information.
  - Complex Contagion: Requires exposure to multiple individuals to spread information
  - Historical Habit: Persistence of previous years' behavior.
- Disease Model: Run a standard SIR (Susceptible-Infected-Recovered) model on the spatial network.
  - Transmissibility rate: 0.05 to 0.5 to reflect varying respiratory pathogens.
  - Recovery rate: 5–7 days (standard influenza parameters).

Who interacts with whom?



How interconnected are groups?

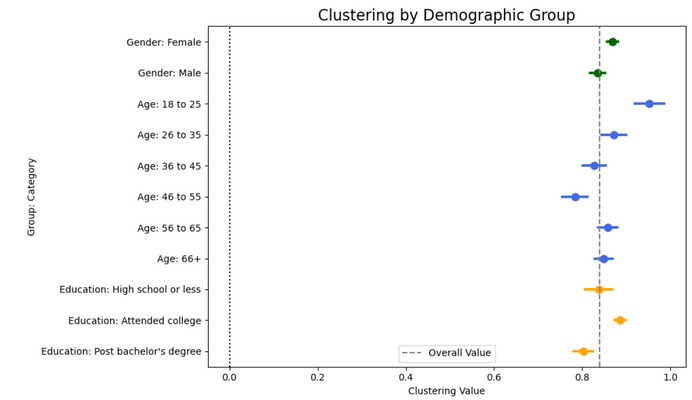


Fig 1. a. Assortativity of contacts by age groups. The positive correlation between groups show that individuals tend to have more contacts within their age group than with other age groups. b. Clustering of contacts by demographic groups gender, age, and education. The dotted line represents zero clustering, the dashed line represents average clustering across all participants. Clustering is quite similar across demographics with the most variation across age groups with younger groups having higher than average clustering.

Simulating prevalence with our model

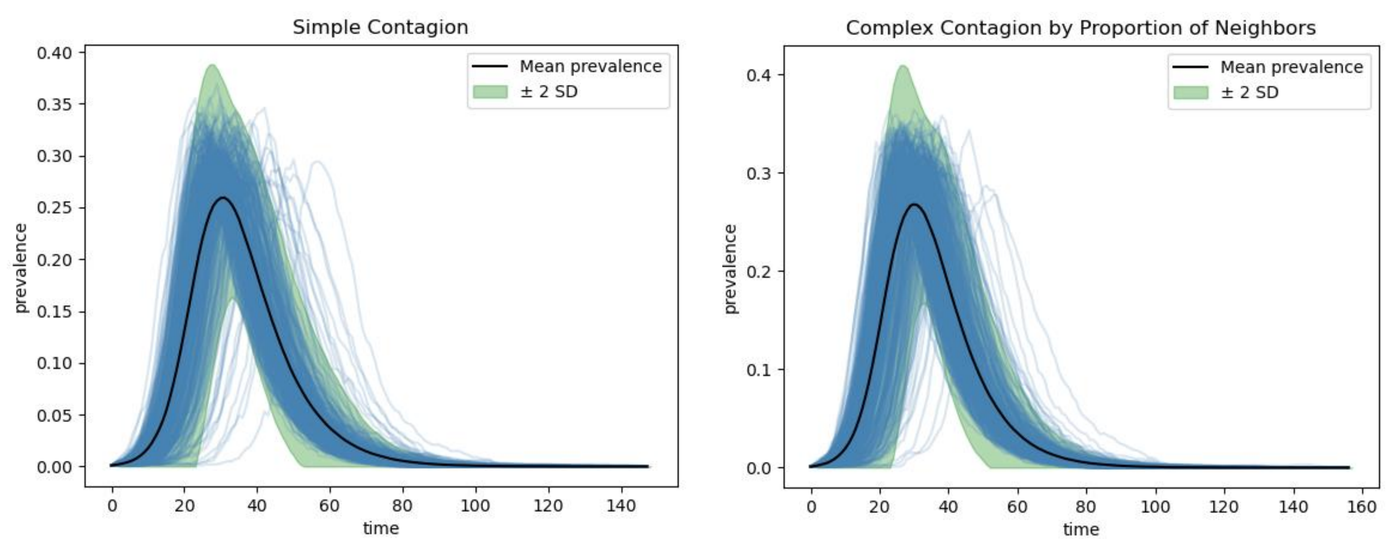


Fig 2.Disease prevalence rates using an ERGM sample with 20 different combinations of both disease and behavioral spread. N = 1000, recovery rate = 0.1, basic reproduction number = 5, vaccination rate = 0.1, complex contagion proportion = 0.7. The mean peak size for simple and complex contagion are 0.2994 and 0.3131 respectively and there is not a significant difference between the two at this low vaccination rate. Future work will analyze how scaling the vaccination, recovery, and transmissibility rate affects the prevalence differently.